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BIOPHYSICS

The effect of microgravity and flows on the organization and dynamics of the human genome

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The human genome is highly organized and dynamic across different length scales. Genome organization and dynamics change upon exposure to mechanical stresses, leading to changes in transcription, replication, and DNA repair. Gravity presents a constant mechanical stress applied to the genome—in its absence, i.e., microgravity, gene expression markedly changes—yet the role of gravity in the genome's organization and function is unknown. Here, we investigate changes in the genome's organization and dynamics upon simulated microgravity and flows. We find that the nucleus changes its shape and size, with liquid phases inside the nucleus, such as nucleolus, visibly affected by the lack of buoyancy and convection in the nucleoplasm. Notably, the genome organization and dynamics remain largely preserved. While flows cause DNA damage, simulated microgravity does not. Our study provides fundamental biophysical insights into the nuclear and genome organization, highlighting its robustness against mechanical stresses and revealing key aspects of the role of gravity in the cell.

INTRODUCTION

The genome contains an essential information for life of a cell, which is encoded in the form of DNA molecules. DNA and associated proteins form a chromatin fiber, which is hierarchically folded into loops, topologically associating domains, A/B compartments, and chromosome territories inside the cell nucleus (*1*–*4*). Chromatin serves as a template for key biological processes, such as transcription and replication (*1*). Timescales and length scales of these processes are directly affected by physical properties of chromatin (e.g., persistence length and packing density), which in turn affect all cellular processes via the central dogma of biology (*5*–*8*). Notably, these physical properties were shown to be altered after nucleus' exposure to mechanical stresses (*9*–*16*). There is one persistent mechanical stress—gravity—in the presence of which all life on Earth has evolved. Yet, the role of gravity in the genome's organization and function remains unknown.

Mechanical stress applied to the nucleus in the form of micropipette aspiration, micromanipulation, microinjection, or constricted migration led to chromatin becoming more viscous (*11*, *13*–*17*). This is largely due to softening of the transcriptionally inactive heterochromatin, because of the loss of B compartment interaction strength (*13*–*15*, *17*). In addition, up-regulation of transcription factors related to adhesion (SRF) and mechanotransduction (YAP) has been linked to compression and shear stresses applied to the cell (*12*, *18*, *19*). This affects the cell adhesion and tissue formation, and was proposed to be sensitive to gravity (*18*, *19*). However, to truly reveal the effect of gravity, one must remove it from the system of interest and observe its subsequent response.

The effects of zero gravity (or microgravity) can be studied in space or simulated on Earth. The microgravity has been shown to cause effects ranging from the level of individual cells, organs to human body (*20*, *21*). The microgravity environment affects individual cells (~10 μm) despite the fact that the magnitude of gravitational force on a single cell is ~0.01 pN and thus dwarfed by other intracellular forces, such as the pushing force by individual microtubules (~3 pN) or the force needed to detach a single integrin link to the actin network (1 to 10 pN) (*22*, *23*, *24*). Large cells such as *Xenopus laevis* oocytes (~500 μm) use an F-actin scaffold to suspend their subnuclear bodies (e.g., nucleoli) against gravity (*25*). Such a direct link to gravity has not been shown in smaller cells; however, it has been speculated that animal cells may sense gravity indirectly through a series of stress-dependent changes in their internal structure and extracellular matrix, transmitting these forces to the genome via mechanotransduction (*26*, *27*).

The human genome was shown to have a marked response to microgravity in space, with increased chromosomal abnormalities, persistent telomere loss, and gene expression changes (*20*). Differential gene expression included transcription factors regulating histone modifications, which could affect chromatin structure and function (*28*). While gene expression changes have been firmly established, a physical effect of lack of gravitational forces on genome organization remains unknown. Considering that nuclear organization is upheld by liquid-liquid phase separations of the nucleoplasm (*29*–*33*), a question arises how phase separations are maintained in the absence of gravity. Formation of liquid droplets by phase separation relies on buoyancy-driven flows, which are nonexistent in microgravity (*34*–*36*). Thus, microgravity may affect the nuclear organization and, by extension, the genome.

While outer space is the ideal place to study effects of microgravity, such studies are confounded by the effects of space-borne radiation on cells (*37*). Hence, analogous Earth-based studies that simulate microgravity are essential. To this end, parabolic flights can produce microgravity for ~30 s at a time. In the laboratory, microgravity can be simulated by a clinostat or random positioning machine (RPM), devices that rotate cells in two (2D) and three (3D) dimensions, respectively (*38*–*40*). By rotating samples relative to the gravity vector, clinostat and RPM can achieve microgravity on timescales shorter than studied cellular processes. Experiments with human cells revealed that simulated microgravity led to changes in cell adhesion, cytoskeleton organization, nuclear shape, and extracellular matrix (*41*, *42*). At the genomic level, genes related to growth,

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development, and immune response were found to be down-regulated, while some oncogenes (KRAS and VIM) were up-regulated (*43*–*46*). Cell proliferation and differentiation have been reported to increase/decrease across different cell lines and experimental conditions (*47*–*50*). While all of these observations suggest that cells are strongly affected by simulated microgravity, it is important to note that such experiments can easily be confounded by the presence of air bubbles in the sample and/or fluid flows generated upon clinostat/RPM rotation (*51*). A recent study revealed that suppression of myotube formation in a fast-rotating clinostat (60 rpm) is due to fluid motion, not simulated microgravity (*52*). Thus, systematic experimental controls are imperative to uncover the effect of simulated microgravity. Moreover, to date, most microgravity-exposed cells were fixed for either imaging or biochemical analysis. Live-cell imaging during/after microgravity exposure is lagging behind (*53*–*55*), leaving many changes to cell physiology still unexplored.

In this work, we study the effects of simulated gravity and flows on the human genome, its organization and dynamics, in live cells. To this end, we design and build a custom RPM, which can follow any 3D rotational path, thus allowing us to induce flows as well as simulate microgravity. We develop a sample preparation protocol that eliminates air bubbles in the sample and allows for live-cell imaging immediately following the cells' mechanical perturbation. We use a suite of human transgenic cell lines to visualize the genome, nuclear envelope (NE), DNA double-strand breaks (DSBs), and nucleolus in response to flows and simulated microgravity, and carry out quantitative physical analysis of observed changes. We find that while the nucleus changes its shape and size in response to flows and microgravity, the genome is unexpectedly robust. Our data provide fundamental insights into effects of flows and simulated microgravity on the genome's organization and dynamics.

RESULTS

Design and validation of RPM

To study effects of microgravity and flows on the cell nucleus and the genome in live human cells, we design and construct an RPM, which allows for high-resolution live-cell imaging directly after the microgravity exposure (Fig. 1A; see Materials and Methods and figs. S1 and S2). The RPM operates inside a cell incubator (Fig. 1B), thus maintaining optimal cell culture conditions. In contrast to earlier RPMs, which used cell culture flasks (e.g., T25) enabling biochemical assays, such as gene and protein expression (*56*–*58*), our custom RPM is optimized for use of imaging dishes (e.g., 35-mm MatTek dishes; Fig. 1B, inset), thus enabling imaging of live cells immediately after their microgravity exposure.

The main principle of an RPM is the 3D rotation of a sample (*40*). This is achieved by simultaneous vertical and horizontal sample rotation, executed by simultaneous motion of RPM's inner and outer frames (Fig. 1A). Compared to a traditional clinostat, which only rotates samples vertically (*59*), an RPM offers microgravity exposure to all sides of the sample (Fig. 1A, inset). In an RPM, the sample is constantly rotated in 3D, thus over time its normal vector points in all directions. Hence, the angle between the sample's normal vector and gravity averages to 0 over time. The motors driving the motion of RPM inner and outer frames can be controlled to follow any user-generated path.

The choice of an RPM path is not trivial, as some paths may cause extraneous fluid flows, while others lead to heterogeneous coverage

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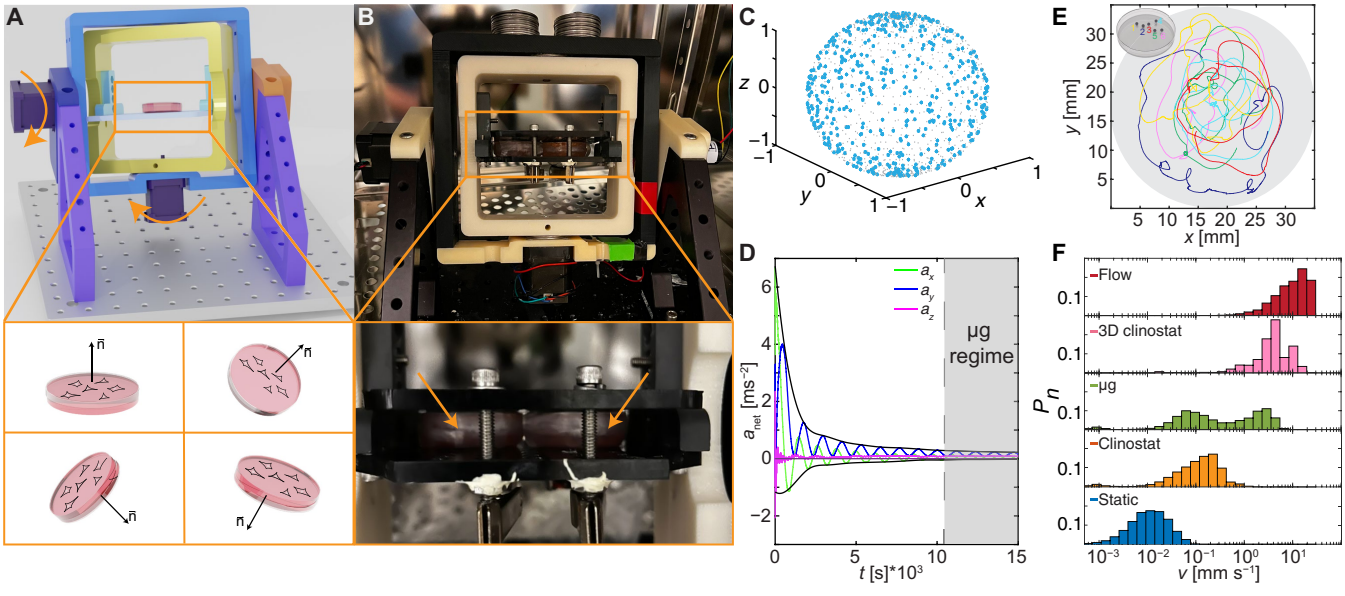


Fig. 1. RPM: Design and path validation. (**A**) Drawing of RPM highlighting individual components. Detailed description provided in fig. S1. Inset: cartoon showing different orientations of cell culture dish during RPM rotation, displaying its normal vector $\vec{n}$. (**B**) Image of RPM loaded with samples in the cell incubator. Inset: two cell culture dishes loaded on the RPM. (**C**) 3D coverage of the normal vector over 5 hours of microgravity exposure. (**D**) Net acceleration, a_{net}, computed from measured components a_x, a_y, a_z in *x*, *y*, and *z* directions over 4 hours on the microgravity path. (**E**) Example bead trajectories during exposure to microgravity. The measured bead motion is relative to the RPM rotation (camera moves with RPM). Inset: cartoon of a cell culture dish filled with fluid and density-matched beads for visualization of fluid flows during RPM rotation. (**F**) Velocity histograms for beads exposed to the following RPM paths: vertical rotation at 42° s^{-1} (Clinostat, yellow), custom microgravity path at an average of 42° s^{-1} (μg, green), both RPM frames rotating at 42° s^{-1} (3D Clinostat, pink), fast horizontal rotation at 800° s^{-1} (Flow, red), and static path (Static, *blue*). Velocities measured over 1-s intervals. Sample sizes, means ± SD, and *P* values are listed in table S1.

of the 3D solid angles (e.g., showing higher likelihood of visiting the poles). Fluid flow inside the sample dish is generated through the 3D sample rotation and is the main confounding factor in simulated microgravity experiments on Earth (*51*, *52*). Microgravity experiments in space involve no rotation (*60*). To minimize the fluid flows in simulated microgravity experiments, the RPM direction changes and speed need to be reduced (fig. S3). An optimal solution is to run the RPM without change in direction or speed; however, such path leads to more frequent visits to the poles of the 3D rotation (fig. S3B). To solve this issue, a linear sawtooth path, during which the RPM accelerates past the poles, was suggested (*61*). In addition, a speed of 42° s^{-1} was shown to minimize the flows (*51*).

To identify an optimal path for microgravity exposure of live human cells, we tested a series of different paths and evaluated their fluid flow generation as well as homogeneous coverage of the 3D solid angle. To this end, we use an accelerometer to measure net RPM acceleration and 1-mm density-matched beads immersed in water in a 35-mm MatTek dish to measure fluid flows (figs. S3 and S4). Specifically, we tested vertical rotation (clinostat mode), two-axis constant rotation (3D clinostat mode), an arc-based randomly rotating path (*62*), and a linear sawtooth smoothly rotating path (*61*). We find the optimal RPM path, which we term as our μg path, by rotating the outer frame with a linear sawtooth pattern at an average speed of 42° s^{-1} and the inner frame rotating at a constant speed of 42° s^{-1}. Using an accelerometer, we find a homogeneous coverage of the 3D solid angle by the sample's normal vector (Fig. 1C and fig. S3D). The μg path achieves microgravity after 3 hours, with a net acceleration vector $\vec{a} = (0.05,0.09,0.07)\vec{g}$ (Fig. 1D), which is comparable with an RPM operating in the 3D clinostat mode (*63*). The net residual acceleration is equivalent to ~0.1*g*, which corresponds to a force of ~0.001 pN on a single cell. During the μg path, both RPM frames complete a full rotation every 8.5 s. Thus, once a microgravity regime is reached, it allows for studies of cellular processes over relevant timescales.

To visualize and assess fluid flows in the sample dish during RPM rotation, we track the motion of density-matched 1-mm beads immersed in water. Figure 1E shows trajectories of beads exposed to the μg path. We measured the bead velocities over 1-s intervals during 120-s exposure to all studied kinematic regimes (table S1). Figure 1F shows the velocity histograms v, which confirm that our chosen paths lead to different v. Together, they cover v over almost five orders of magnitude (10^{-3} to 10^{2} mm s^{-1}), allowing us to systematically examine effects of flows: The static program keeps the sample fixed in the horizontal plane, generating no flow and having no effect on gravity ($\langle v \rangle = 0.02$ mm s^{-1}). The flow program keeps the sample horizontal and applies high-speed rotation around a vertical axis in alternating clockwise and counterclockwise motion and, as expected, generates strong flows with no effect on gravity ($\langle v \rangle = 12$ mm s^{-1}). The remaining three programs aim to cancel gravity through different rotational kinematics that induce different v, with clinostat leading to smallest ($\langle v \rangle = 0.15$ mm s^{-1}) and 3D clinostat to largest flows ($\langle v \rangle = 4.5$ mm s^{-1}). Our μg regime shows a bimodal velocity distribution, with the first and second modes similar to flow velocities in clinostat and 3D clinostat, respectively. Thus, on average, our μg path leads to substantially reduced flows ($\langle v \rangle = 0.99$ mm s^{-1}), when compared to 3D clinostat. This is critical, as the flows of cell media in the sample dish lead to shear stresses on cells, which can affect their physiology (*64*).

To assess the effects of flows in our μg path, we compute the Froude number, a dimensionless number that allows for comparison of the flow and gravitational effects. Specifically, $\mathrm{Fr} = v / \sqrt{gL}$, where v is the flow velocity, g is the gravitational acceleration, and L is the characteristic length scale, here, the cell size. For the flow regime, $v = 12$ mm s^{-1}, $g = 9.81$ m s^{-2}, $L \approx 10$ μm, and for the μg regime, $v = 1$ mm s^{-1}, $g = 1.22$ m s^{-2}, $L \approx 10$ μm. This yields $\mathrm{Fr}_{\mathrm{flow}} = 1.21$ and $\mathrm{Fr}_{\mu g} = 0.29$ and confirms that our flow regime is dominated by flows ($\mathrm{Fr}_{flow} > 1$) and μg regime by the gravity effects ($\mathrm{Fr}_{\mu g} < 1$).

Once we established the optimal μg path for the RPM rotation, we design an RPM experiment for live cells and its proper controls as follows: We prepare a sealed sample dish with live cells adhered at the glass bottom and filled with cell media, which we load into RPM for a 24-hour exposure to the μg path (see Materials and Methods). First, we measure pH and dissolved O_2 levels in the cell media before and after μg exposure to confirm that the lack of gas exchange due to sample dish sealing does not affect the cell growth. We find only a modest reduction in dissolved O_2, from 21 to 14% oxygen by volume (table S2), which is far above the physiological levels of 5% oxygen by volume in cell (*65*). We find a small pH decrease from 7.36 to 6.99 (table S2), which is well within the optimal pH range for HeLa cell growth (*66*).

To correctly identify effects of microgravity on cells and rule out any confounding factors, proper controls are needed (*40*, *52*). We use three different controls: (i) Static: To identify the effect of microgravity on cells, we compare against cells that were not exposed. Namely, a sample dish with adherent cells on its glass bottom and filled with 2 ml of cell media stays in the cell incubator without sealing and microgravity exposure. (ii) Filled: To check whether the sample sealing affects the cells, we prepare a control sample as in (i) but completely filled with cell media and sealed. (iii) Flow: To identify the effect of flows, we expose a sealed sample dish with cells to the horizontal RPM rotation with changing direction, which we found to cause the highest flow in the sample (Fig. 1F).

Figure 2 shows micrographs of cells under all four studied conditions: μg, static, filled, and flow. When compared to the static control (Fig. 2A), we find no visible changes in cell morphology or growth for the filled control (Fig. 2B) or the μg sample (Fig. 2C). In contrast, in the flow condition, the cells appear more elongated and spread out than in any other condition (Fig. 2D). We find that cells

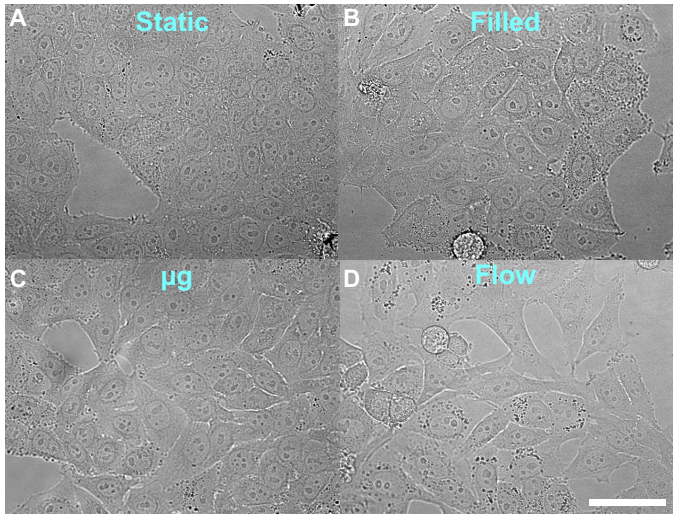


Fig. 2. Cells exposed to microgravity and flows in RPM. Micrographs of cells under (**A**) static, (**B**) filled, (**C**) microgravity (μg), and (**D**) flow conditions. Scale bar, 50 μm.

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remain adhered under all conditions; however, we observe a substantial number of detached single cells in the flow condition. We observe no multicellular spheroid (MCS) formation, a common occurrence in many earlier microgravity experiments (*63*, *67*, *68*), across all our samples. The reason for this improvement may be the air bubble–free environment achieved in our sample preparation (see Materials and Methods) combined with long cell adherence time (24 hours). The presence of air bubbles in samples has recently been found to cause increased cell detachment during RPM rotation, while an increased cell adherence time before the microgravity experiment was found to reduce spheroid formation (*58*). Thus, the combination of long cell adherence times, air bubble–free fluid and slow rotation speed (42° s^{-1}) may prevent formation of MCSs, demonstrating low cellular stress in our experiments.

Taken together, we identified optimal conditions, including sample preparation and 3D RPM rotation path, to simulate microgravity for live human cells. Our custom RPM allows for immediate high-resolution live-cell imaging after microgravity exposure. Next, we will use this experimental system and approach to investigate effects of microgravity and flows on the human cell nucleus and the genome. For space and clarity reasons, we present all data for the filled condition in the Supplementary Materials, as we found it to be equivalent to the static control in all measurements.

Effect of microgravity on the cell nucleus

To uncover the effects of microgravity on the human cell nucleus, we monitored its size and shape upon μg exposure. Using spinning disk confocal microscopy, we imaged transgenic HeLa H2B-GFP cells to visualize nuclei (Fig. 3). Specifically, we conduct low-magnification (40×) imaging surveys of large cell numbers, with 5214, 1343, and 1203 nuclei in the static, μg, and flow conditions, respectively (Fig. 3A), and evaluate the nuclear area A_{Nucleus} and eccentricity e_{Nucleus} of every nucleus (for filled condition, see fig. S5). e_{Nucleus} is computed as the ratio of the semimajor to the semiminor axis length, with 1 corresponding to a circle. We find that A_{Nucleus} significantly increases in both μg (by ~14%) and flow (by ~27%) conditions ($P < 10^{-3}$) and note that e_{Nucleus} increases significantly (by ~13%) only in the flow condition (Fig. 3, B and C). Means ± SD and *P* values for A_{Nucleus} and e_{Nucleus} are listed in table S3. This analysis was carried out for HeLa H2B-GFP cells with three different types of secondary expression (LMNA-mCherry/53BP1-mCherry/NPM-mApple) used in this study. We pooled these data into a single population, after we found A_{Nucleus} and e_{Nucleus} distributions of these three populations to be equivalent (fig. S6 and table S3). The large change of A_{Nucleus} in the flow condition is likely due to high detachment of the cells, skewing the distribution towards large adherent cells.

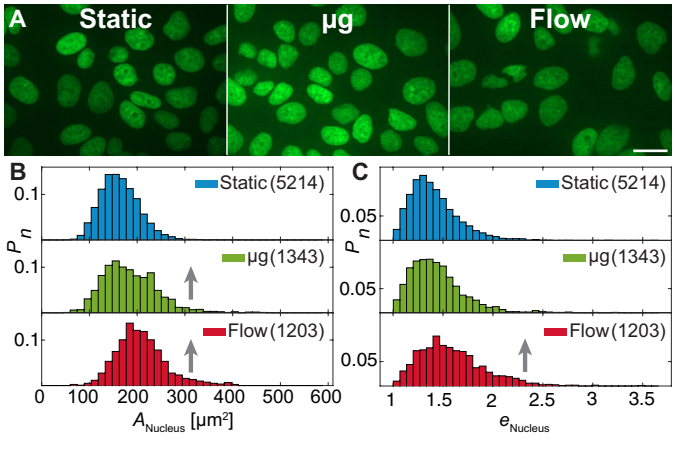


Fig. 3. Effect of microgravity and flows on the cell nucleus. Micrographs of nuclei expressing H2B-GFP under (**A**) static, microgravity exposure (μg), and flow conditions. (**B** and **C**) Distributions of nuclear area A_{nucleus} and eccentricity e_{Nucleus}, obtained from H2B-GFP signal (40× magnification). *N* in the legend denotes the number of nuclei analyzed. Means ± SD and *P* values are listed in table S3. Scale bar, 20 μm.

To reveal any subtle changes in the nuclear shape, we collected imaging data of HeLa H2B-GFP/LMNA-mCherry cells at high (100×) magnification (Fig. 4, A and B). These cells allow us to assess both chromatin (H2B-GFP) and NE (LMNA-mCherry) under static, μg, and flow conditions (for filled condition, see fig. S7). To measure the height, h_{Nucleus}, and volume, V_{Nucleus}, of the nucleus, we collect *z*-stacks with 0.5-μm step size of LMNA-mCherry signal from 98, 70, and 55 nuclei in the static, μg, and flow conditions (fig. S8). Our data show an increase in V_{Nucleus} in μg (by ~18%) and flow (by ~15%) conditions (Fig. 4C, $P < 10^{-3}$). Furthermore, we find that h_{Nucleus} decreases as A_{Nucleus} increases (Fig. 4D), with Pearson coefficients of −0.542 for static, −0.237 for μg, and −0.335 for flow conditions ($P < 0.05$ for all conditions), but we measure no significant change in h_{Nucleus} in any condition (fig. S8 and table S4).

To assess whether any changes occur in the NE structure upon microgravity exposure, we examine the LMNA-mCherry signal as follows: We first account for different expression levels of LMNA-mCherry across cells by computing the normalized intensity $I_{n,\mathrm{Lamin}} = \frac{I - I_{\min}}{I_{\mathrm{avg}}}$ for every pixel along the nuclear contour, where I, $I_{\min}$, and I_{avg} are the raw, minimal, and average pixel intensity for the given nucleus. Then, we compute $\langle I_{n,\mathrm{Lamin}} \rangle$ for all nuclei and find no changes under any of the studied conditions (fig. S7 and table S5). In addition, we evaluate the average NE thickness, $\langle d_{\mathrm{Lamin}} \rangle$, in the focal plane of the nucleus. We find that $\langle d_{\mathrm{Lamin}} \rangle \approx 0.3 \pm 0.02$ μm remains unchanged across all studied conditions (Fig. 4E, fig. S7, and table S5).

Last, we examine whether any changes occur in the NE dynamics upon microgravity exposure. To this end, we calculate the NE fluctuations as $u(\phi, t) = r(\phi, t) - r_0(\phi)$, where u denotes the deviation of the instantaneous contour $r(\phi, t)$ from the average contour $r_0(\phi)$ at time t (*69*). Figure 4F shows an example of the NE contour fluctuating over time, while Fig. 4G displays u^2, the squared magnitude of these fluctuations. To determine the noise floor of this measurement, we evaluated NE fluctuations in the nuclei fixed in formaldehyde (Fig. 4G, black dot) and confirmed that all our measurements are well above the noise floor. Further, we measured the q-dependent NE fluctuations, by performing the Fourier transformation of $u(\phi, t)$ at every time t, following $u_q(t) = \frac{1}{2}\pi \int_0^{2\pi} u(\phi, t) e^{-iq\phi}\, d\phi$, and calculating the mean square amplitude $\langle u_q^2 \rangle$ as previously described (*69*, *70*). Figure 4H shows $\langle u_q^2 \rangle$ as a function of the wave number q averaged over 150, 44, and 39 nuclei in the static, μg, and flow conditions, respectively. Again, NE fluctuations of nuclei fixed in formaldehyde serve as the noise floor measurement. Our data for the static control are in agreement with our previous study (*69*). We find the NE dynamics ($\langle u_q^2 \rangle$) to be largely unchanged upon microgravity or flow exposure, with an exception of an increase of ~30% at high q values, when exposed to microgravity ($P < 0.02$; fig. S7D and table S6).

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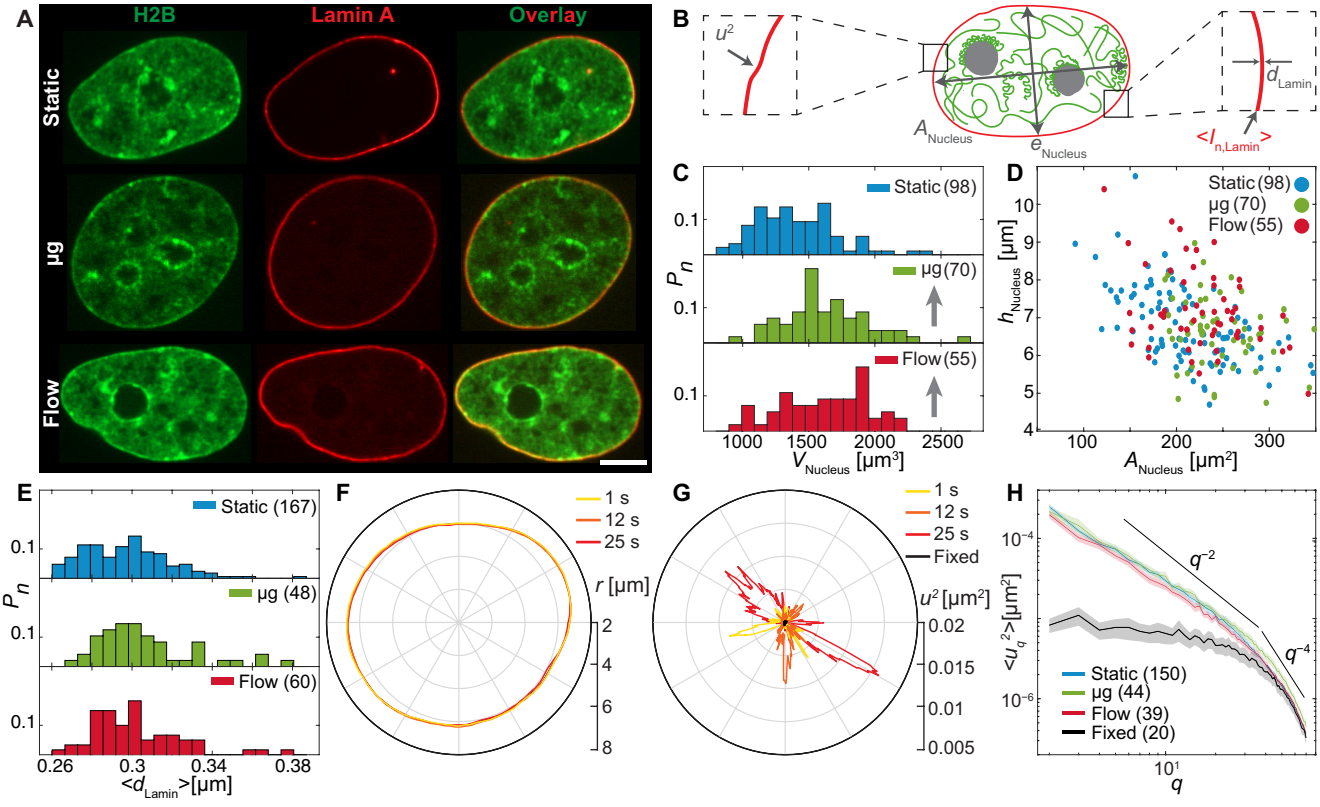


Fig. 4. Effect of microgravity and flows on the cell nucleus. (**A**) Micrographs of nuclei with fluorescently labeled chromatin (green, H2B-GFP) and NE (red, LMNA-mCherry) under static, microgravity (μg), and fluid flow conditions. (**B**) Cartoon illustrating the evaluated physical properties of the nucleus. (**C**) Distributions of nuclear volume $V_{Nucleus}$. (**D**) Scatterplot of nuclear height $h_{Nucleus}$ versus $A_{Nucleus}$. (**E**) Distribution of average NE thickness $\langle d_{Lamin} \rangle$. (**F**) Contour of the NE (LMNA-mCherry) at $t = 1$, 5, and 12 s in the static condition. (**G**) NE fluctuations, u^2, at $t = 1$, 5, and 12 s obtained for contour in (F). Black dot at origin shows NE fluctuations of a nucleus fixed with formaldehyde, confirming that all our u^2 measurements are above the noise floor. (**H**) Wave number–dependent fluctuations $\left\langle u_q^2 \right\rangle$ for the NE under static, μg, and flow conditions. Black line represents $\left\langle u_q^2 \right\rangle$ measured for nuclei fixed with formaldehyde. *N* in the legend denotes a number of nuclei analyzed. Means ± SD and *P* values are listed in tables S4 to S6. Scale bar, 5 μm.

Since NE fluctuations were shown to be actively driven by actin and microtubule polymerization (*69*), we surveyed changes to actin and microtubule intracellular organization upon microgravity exposure. To this end, we visualized actin (Phalloidin-Alexa Fluor 405) and microtubules (αTubulin-Alexa Fluor 647) in cells under static, μg, and flow conditions (fig. S9). We note anomalous actin filament arrangements, including an increase in punctate accumulations of actin filaments in the μg-exposed cells, which is in agreement with previous studies (*71*). While we do not see any strong qualitative changes in the microtubules of μg-exposed cells, we cannot rule out more subtle effects. For example, past work showed disruption of mitotic spindles and shorter, wavier, and more fragmented interphase microtubules upon μg exposure (*72*, *73*). Indeed, in the μg condition, we find cells to be smaller and packed tighter together, exhibiting a smaller cytoplasm, which would be consistent with shorter microtubules. In the flow condition, we see isolated cells with long cytoplasmic protrusions and an increased presence of actin fibers that span the entire cytoplasm but no changes in the microtubule organization. Overall, our data show that the nucleus changes its size and shape upon flows and μg exposure, while the NE structure and dynamics remain largely unchanged.

Response of the genome to simulated microgravity

Next, we investigate how simulated microgravity affects the human genome. In particular, we focus on its organization and dynamics, both of which are critical for its proper function (*2*–*4*, *7*). To evaluate changes in the genome organization, we examine the chromatin density distribution across the nucleus under different conditions. To this end, we use H2B-GFP signal as a proxy for chromatin density (*74*). For every pixel in the nucleus, we compute the normalized H2B-GFP intensity $I_n = \frac{I - I_{min}}{I_{avg}}$, where I, I_{min}, and I_{avg} are the raw, minimal, and average pixel intensity for the given nucleus. We then compute average $\langle I_n \rangle$ for each nucleus. When we evaluate $\langle I_n \rangle$ distributions of 830, 252, and 254 nuclei in the static, μg, and flow conditions, respectively, we find no significant changes (Fig. 5A; for filled condition, see fig. S10). We compute the standard deviation (SD) σ_I of the I_n distribution for each nucleus, which reflects the variation of density states. We find that σ_I changes by less than 5% in μg and flow conditions (Fig. 5B and table S7). Thus, our data suggest that intranuclear chromatin density distribution remains largely unchanged upon cell exposure to μg or flows.

To evaluate genomic motions across the entire nucleus, we perform displacement correlation spectroscopy [DCS; (*75*)]. We record

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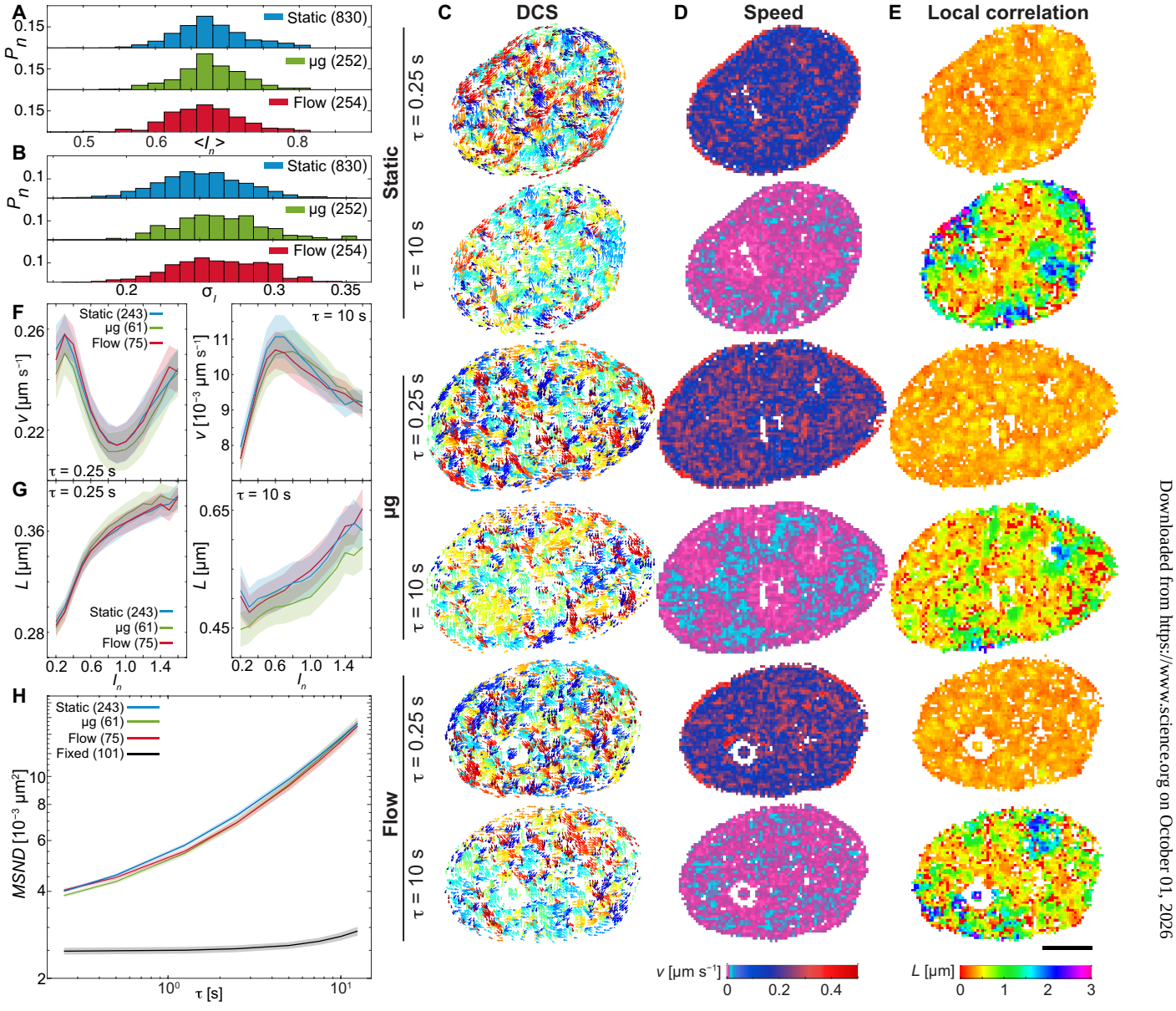


Fig. 5. The genome upon microgravity and flow exposure. (**A**) Distribution of $\langle I_n \rangle$ over nuclei populations, where $\langle I_n \rangle$ is the average normalized H2B-GFP intensity across a nucleus. (**B**) Distribution of SD, σ_I, measured for I_n distribution of each nucleus. (**C**) Displacement fields $\vec{d}(\vec{r}, \tau)$ for nuclei under static, μg, and flow conditions at $\tau = 0.25$ and 10 s. (**D**) Heatmaps of average local chromatin speed of chromatin $\langle v(\vec{r}) \rangle$ for nuclei from (C) at $\tau = 0.25$ and 10 s averaged over all time points. (**E**) Heatmaps of average local correlation length L of chromatin displacements for nuclei from (C) at $\tau = 0.25$ and 10 s averaged over all time points. (**F**) v as a function of the chromatin compaction I_n at $\tau = 0.25$ and 10 s, averaged over all time points. (**G**) L as a function of I_n at $\tau = 0.25$ and 10 s, averaged over all time points. (**H**) The average *MSND* for nuclei populations under static, μg, and flow conditions. Error bars denote SE. *N* in the legend denotes number of nuclei analyzed. Means ± SD and *P* values are listed in tables S7 to S9. Nuclei rotated for optimal figure spacing. Scale bar, 5 μm.

high-resolution imaging streams of nuclei with fluorescently labeled chromatin (H2B-GFP) with a temporal resolution of 250 ms and a duration of 25 s, followed by a DCS analysis of 243, 61, and 75 nuclei in the static, μg, and flow conditions, respectively. Using DCS, we survey chromatin dynamics across different timescales. Figure 5C shows displacement fields $\vec{d}(\vec{r}, t)$ at $\tau = 0.25$ and 10 s for nuclei under different conditions, with displacement vectors color-coded by their direction. We confirm that chromatin moves coherently at longer times ($\tau = 10$ s) for all studied conditions (fig. S11). In addition, we review at each $\vec{r}$ the local chromatin speed $v(\vec{r}) = |\vec{d}(\vec{r})/\tau|$ (Fig. 5D) and local spatial correlation of chromatin displacements $C_L(\vec{r}, \Delta\vec{r}) = \langle \vec{d}(\vec{r}, \Delta t), \vec{d}(\vec{r} + \Delta\vec{r}, \Delta t) \rangle$ at every $\vec{r}$ for $\tau = 0.25$ and 10 s. We then fit the radially and time-averaged $\langle C_L(\vec{r}, \Delta\vec{r}) \rangle$ to an exponential decay providing us the local correlation length $L(\vec{r})$ (Fig. 5E). Distributions of v and L across all nuclei at $\tau = 0.25$ and 10 s are shown in fig. S12 (table S8). We find no significant changes for v or L at either timescale in any condition. Last, we analyze v and L as a function of chromatin compaction I_n (Fig. 5, F and G). We

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find L to increase with I_n as previously described for differentiated cells (*76*).

To further assess changes in genomic motions in simulated microgravity, we evaluate the mean squared network displacement, $MSND(\tau) = \left\langle |\vec{d}(\vec{r},\tau)|^2 \right\rangle$, where $\vec{d}(\vec{r},\tau)$ denotes chromatin displacements at position $\vec{r}$. Figure 5H shows the average *MSND* computed over all nuclei in the static (blue line), μg (green line), and flow (red line) conditions (for filled condition, see fig. S12 and table S9). To determine the noise floor of the measurement, we obtain *MSND* for 101 formaldehyde-fixed nuclei and confirm that all our measurements are above the noise floor. Further, we fit the average *MSND*s to $f(\tau) = A + B\tau^{\alpha}$ and find A, B, and α as follows: $A = 0.0028\ \mu m^2$, $B = 0.0026\ \mu m^2\ s^{-\alpha}$, and $\alpha = 0.61$ for the static; $A = 0.0028\ \mu m^2$, $B = 0.0023\ \mu m^2\ s^{-\alpha}$, and $\alpha = 0.66$ for the μg; and $A = 0.0030\ \mu m^2$, $B = 0.0023\ \mu m^2\ s^{-\alpha}$, and $\alpha = 0.65$ for the flow condition, showing no significant changes under any conditions. Overall, our data suggest that the genome, its spatial organization, and dynamics are robust against the mechanical perturbations of simulated microgravity and flows over the studied timescales (24 hours).

DNA damage upon genome's microgravity exposure

The effects of microgravity on cells in space are confounded by radiation damage due to cosmic rays (*37*, *77*). In particular, DNA damage can occur both due to radiation and mechanical stress (*16*, *78*). We evaluate whether RPM-simulated microgravity can cause mechanically induced DNA damage to the human genome. To do so, we expose to microgravity HeLa cells with fluorescently labeled chromatin (H2B-GFP) and DNA damage marker 53BP1, which localizes to DNA DSBs (53BP1-mCherry) (*79*, *80*). Figure 6A shows micrographs of nuclei in static, μg, and flow conditions, as well as nuclei upon neocarzinostatin (NCS) treatment, which chemically induces DSBs (for filled condition, see fig. S13) (*81*, *82*). To confirm that 53BP1-mCherry signal under studied conditions demarks DSBs, we carried out an immunofluorescent staining of γH2AX (Alexa Fluor 405), which is another molecular marker for DSBs (*83*). Figure S13 shows the colocalization of 53BP1-mCherry and γH2AX-Alexa Fluor 405 signals for all conditions, confirming the presence of DSBs.

To assess the extent of DNA damage upon exposure to simulated microgravity, we count the number of DSBs, N_{DSB}, and measure their density, $\rho_{DSB} = N_{DSB} / A_{Nucleus}$, radius, R_{DSB}, distance from the NE, $d_{e,DSB}$, and chromatin compaction at each DSB site, $I_{n,DSB}$, under all studied conditions (Fig. 6B). We find ρ_{DSB} of 0.055 ± 0.040 DSB/μm^2, 0.063 ± 0.033 DSB/μm^2, and 0.071 ± 0.024 DSB/μm^2 for static, μg, and flow conditions, respectively (Fig. 6C and table S10). Our data show an almost 30% increase in ρ_{DSB} under flow conditions ($P < 10^{-3}$), while there is no statistically significant change upon μg. For NCS-treated nuclei, as expected, we see a marked increase in ρ_{DSB} to 0.141 ± 0.063 DSB/μm^2. For R_{DSB}, we find the average DSB radius of ~0.34 μm for static, μg, and flow conditions. For NCS-induced DSBs,

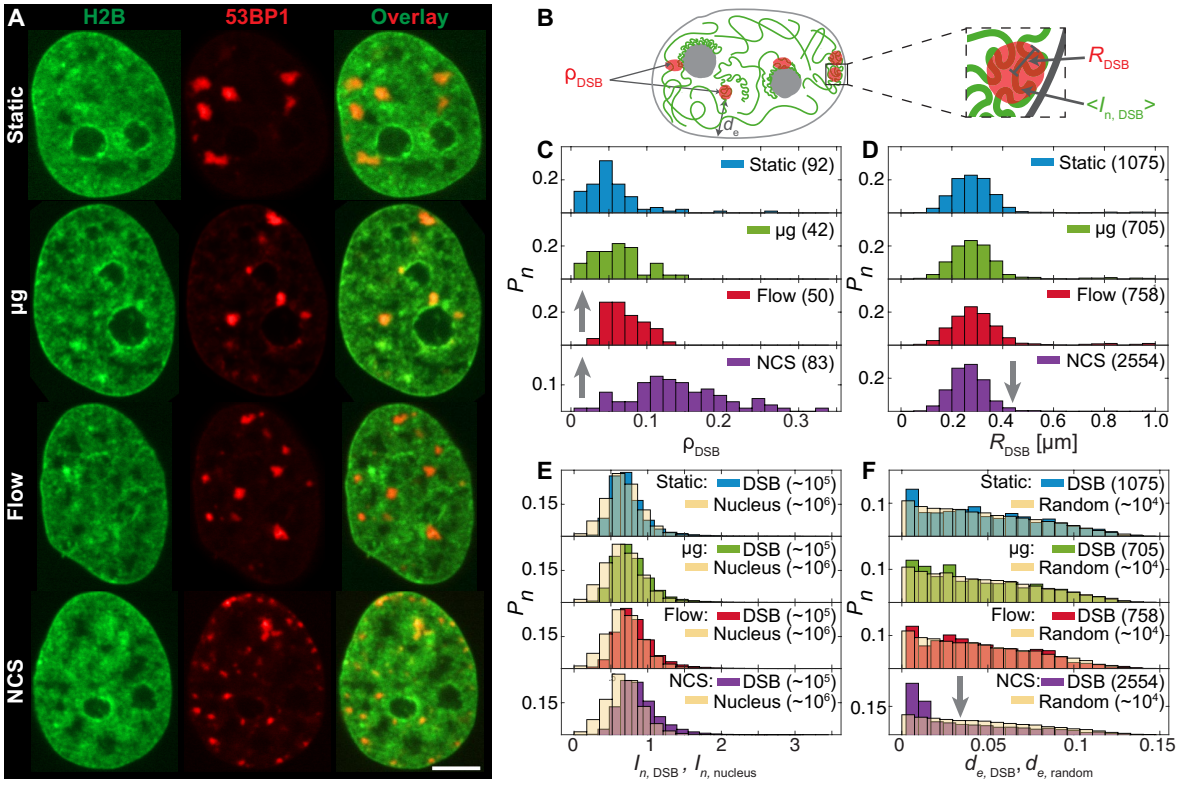


Fig. 6. DNA damage upon microgravity and flow exposure. (**A**) Micrographs of nuclei with fluorescently labeled chromatin (H2B-GFP) and DNA DSBs (53BP1-mCherry) under static, μg, flow, and NCS conditions. (**B**) Cartoon illustrating the measured physical quantities of DSBs. (**C**) Distribution of DSB density, ρ_{DSB}. (**D**) Distribution of DSB radius R_{DSB}. (**E**) Distribution of chromatin compaction within DSB $I_{n,DSB}$ compared to chromatin compaction nucleus-wide $I_{n,nucleus}$. (**F**) Distribution of DSB distances from NE $d_{e,DSB}$, compared to random nuclear sites $d_{e,random}$. Distributions shown for static (blue), μg (green), flow (red), and NCS (purple) conditions. *N* in the legend denotes the number of nuclei (C), DSBs [(D) and (F)], and pixels [(E) and (F)]. Means ± SD and *P* values are shown in table S10. Scale bar, 5 μm.

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we find $R_{DSB} = 0.27 \pm 0.10$ μm, which is in agreement with previous studies (Fig. 6D) (*82*, *84*).

To evaluate local chromatin compaction at DSB sites under different conditions, we compute the normalized H2B-GFP pixel intensity, $I_{n,DSB} = \frac{I_{min} - I}{I_{avg}}$ for all pixels within a region outlined by the 53BP1-mCherry signal of a DSB. We then compare $I_{n,DSB}$ distribution collected over all DSBs from all nuclei against the distribution of all normalized H2B-GFP pixel intensities from all nuclei $I_{n,nucleus}$ (Fig. 6E). We find that $I_{n,DSB}$ distributions in all conditions are shifted toward higher values when compared to $I_{n,nucleus}$ distribution, suggesting more compact chromatin at DSBs. We measure the following means ± SD for the $I_{n,DSB}$ distributions: 0.742 ± 0.265 for static, 0.792 ± 0.275 for μg, 0.829 ± 0.277 for flow, and 0.879 ± 0.275 for the NCS condition. For $I_{n,nucleus}$ distributions, we obtain 0.675 ± 0.253 for static, 0.673 ± 0.264 for μg, 0.694 ± 0.266 for the flow, and 0.682 ± 0.252 for the NCS condition ($P < 10^{-3}$). Last, we measure the DSB distance from the NE, normalized by the nuclear perimeter, $d_{e,DSB}$ for all conditions. We measure the following means ± SD for the $d_{e,DSB}$: 0.049 ± 0.033 for the static, 0.046 ± 0.032 for the μg, 0.046 ± 0.030 for the flow, and 0.035 ± 0.032 for the NCS condition. We compare these values to distances measured for 200 random sites in the nucleus $d_{e,random}$ (Fig. 6F). We find that in all conditions, $d_{e,DSB}$ distributions are comparable to $d_{e,random}$ distribution, except for the NCS condition which is shifted to smaller values ($P < 10^{-3}$). We note that NCS induces many DSBs close to the NE (Fig. 6A), consistent with its physical entry into the nucleus. Overall, we find that simulated microgravity causes no substantial DNA damage, while cells exposed to flows show a large increase in DNA damage.

Response of nucleoli to microgravity

The genome is immersed in the nucleoplasm, which forms many liquid condensates via liquid-liquid phase separation inside the cell nucleus (*32*, *85*). Liquid condensates differ in their density from the surrounding fluid and thus may be uniquely sensitive to microgravity and flows. To examine this hypothesis, we focus on the largest liquid condensate in the nucleus, the nucleolus. Figure 7A shows micrographs of nuclei with fluorescently labeled chromatin (H2B-GFP) and nucleoli (NPM-mApple) under static, μg, and flow conditions (for filled condition, see fig. S14). We do not note any obvious changes in nucleolar phenotype across any condition. To assess any subtle changes in the nucleolar morphology and number, we measure the nucleolar area, A_N, nucleolar eccentricity, e_N, number of negative curvature sites along the nucleolar contour, N_{neg}, the fraction of negative curvature sites, f_{neg}, and the number of nucleoli per nucleus, N_N (Fig. 7B). We carry out these measurements for 503, 138, and 145 nucleoli in static, μg, and flow conditions, respectively.

While we find a 30% increase in A_N for the flow condition ($P < 10^{-3}$), there is no significant change for the μg condition (Fig. 7C). e_N remains unchanged across all conditions (Fig. 7D). We observe a positive correlation between A_N and $A_{Nucleus}$ for all conditions (Fig. 7E), consistent with nucleolar coarsening during the cell cycle (*86*). This is further consistent with the average nucleolar size, $\langle A_N \rangle$, increasing with a decreasing number of nucleoli, N_N, inside a nucleus (Fig. 7F). To examine the nucleolar surface roughness, we count the number of negative curvature sites along the nucleolar contour, N_{neg}, and compute the fraction of negative curvature sites, f_{neg}, following previously published procedures (*87*). We find a significant decrease (~20%) in N_{neg} in both μg and flow conditions ($P < 0.01$; Fig. 7G) and f_{neg} ($P < 0.01$; fig. S14), indicating that nucleoli are rounder and smoother.

Next, we analyze the changes in nucleolar surface tension γ under studied conditions. To do so, we first calculate the nucleolar contour fluctuations u^2 following previously published algorithms (*32*). Figure 7H presents an example of a nucleolar contour fluctuating over time in the static condition, with its fluctuations u^2 plotted in Fig. 7I. Formaldehyde-fixed nucleoli were analyzed to identify the noise floor of the measurement (Fig. 7I, black curve). To estimate the nucleolar surface tension, we assume that nucleolar shape fluctuations are thermally driven (*32*), i.e., $\gamma = k_B T / \langle u^2 \rangle$. We average all u^2 above the noise floor over time and polar angle to obtain $\langle u^2 \rangle$ for 190, 73, and 102 nucleoli in the static, μg, and flow conditions. We find γ to be similar across all conditions, with 1.56 ± 0.312 μN m^{-1} for static, 1.52 ± 0.322 μN m^{-1} for μg, and 1.60 ± 0.336 μN m^{-1} for flow conditions (Fig. 7J and table S11), all of which agree with previous measurements for γ under physiological conditions (*32*). We also examine the chromatin surrounding nucleoli, in particular its compaction. To this end, we compute the normalized H2B-GFP intensity $I_{n,N}$ from the H2B-GFP signal in the region 5 pixels inside and outside of the nucleolar rim as defined by the NPM-mApple signal. Figure S14 shows the average $\langle I_{n,N} \rangle$ to remain unchanged for all conditions. However, its SD, σ_I, shows an increase of 7% for μg and 11% for flow conditions ($P < 0.01$; fig. S14).

Last, nucleolar shape and structure are known to be dependent on RNA Pol I activity, with nucleolar caps forming upon RNA Pol I inhibition (*88*). To investigate whether the changes we observe for nucleoli under some conditions are due to changes in transcriptional activity, we simultaneously visualize chromatin (H2B-GFP), nucleoli (NPM-mApple), RNA Pol I (RPA-Alexa Fluor 405), and RNA Pol II (CTD phosphorylated Ser^5 RNA Pol II-Alexa Fluor 647) (fig. S15). We do not find formation of nucleolar caps under any of the studied conditions, suggesting that RNA Pol I activity remains unperturbed. Similarly, we survey changes in RNA Pol II activity across the nucleus. To this end, we measure the density of active RNA Pol II foci in the nucleus, $\rho_{pol\,II}$, for 204, 131, and 110 nuclei in the static, μg, and flow conditions (fig. S15). We obtain the following means ± SD for the $\rho_{pol\,II}$ distributions: 0.935 ± 0.066 foci/μm^2 for static, 0.942 ± 0.066 foci/μm^2 for μg, and 0.989 ± 0.068 foci/μm^2 for the flow condition. We find no statistically significant changes in ρ_{polII} for the μg condition and a small 6% increase in $\rho_{pol\,II}$ for the flow condition ($P < 10^{-3}$; fig. S15 and table S12). Overall, our data show that the nucleolus changes its size upon flow, but not μg exposure. Nucleolar surface becomes smoother, while its surface tension remains unchanged in response to both perturbations.

DISCUSSION

The goal of our study was to identify the role of gravity in the organization and dynamics of the human genome. To this end, we studied effects of mechanical perturbations that simulate microgravity as well as flows on the cell nucleus. To achieve that, we constructed a custom RPM, which allows for generating both simulated microgravity and controlled flows, and developed sample preparation protocols, which allow for live-cell imaging immediately after cells' exposure to these mechanical perturbations. By employing quantitative imaging and physical analysis, we revealed new biophysical insights into the nucleus, genome, and nucleoplasm.

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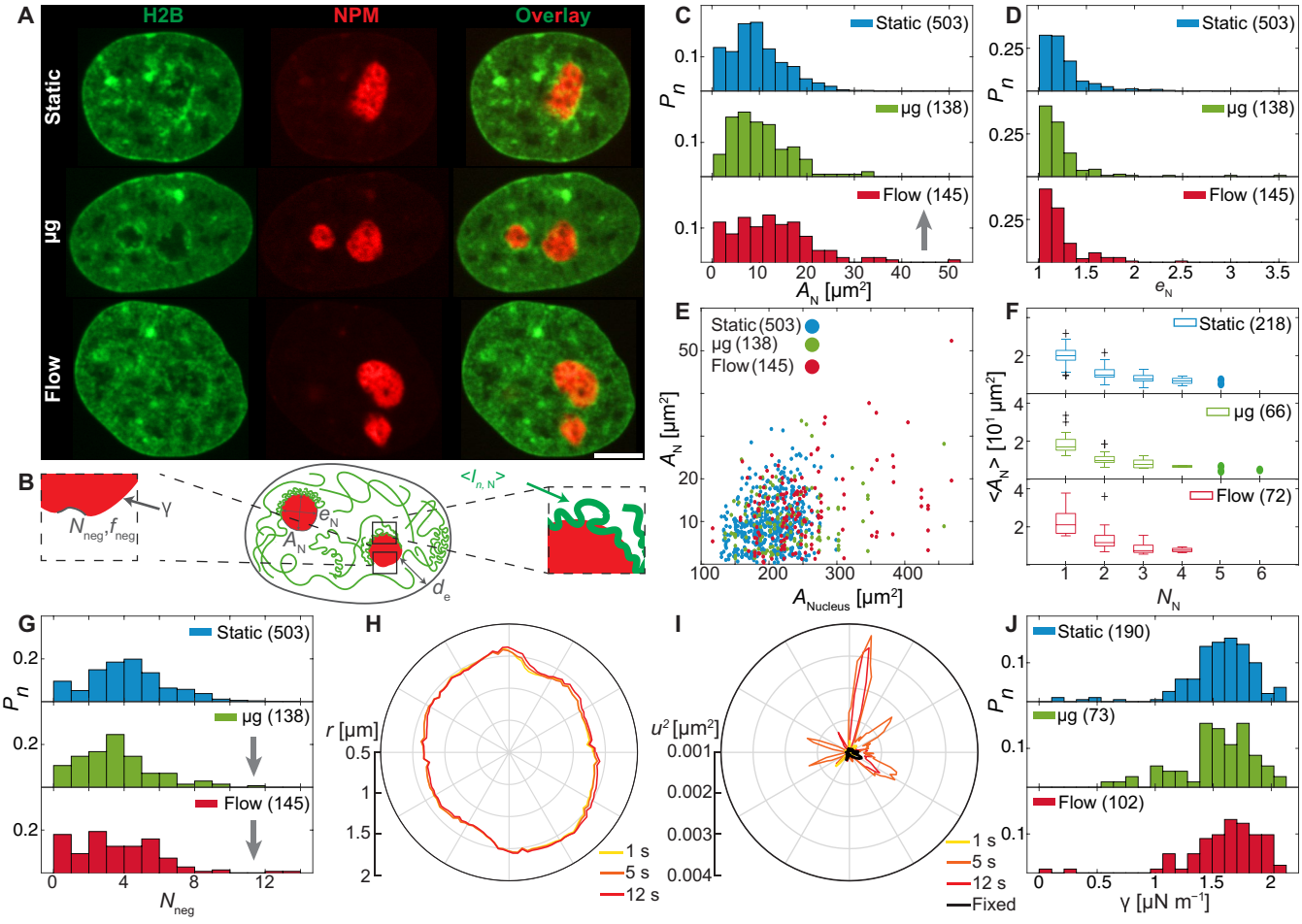


Fig. 7. Response of nucleoli to microgravity and flows. (**A**) Micrographs of nuclei with fluorescently labeled chromatin (H2B-GFP) and nucleoli (NPM-mApple) under static, μg, and flow conditions. (**B**) Cartoon illustrating measured properties of nucleoli. (**C**) Distribution of nucleolar area A_N. (**D**) Distribution of nucleolar eccentricity e_N. (**E**) Scatterplot of nucleolar area A_N versus nuclear area $A_{Nucleus}$. (**F**) Box plots of $\langle A_N \rangle$ versus N_N. (**G**) Distribution of negative curvature sites N_{neg} of the nucleolar contour. (**H**) Nucleolar contour in the static condition at $t = 1$, 5, and 12 s. (**I**) Nucleolar surface fluctuations u^2 for a nucleolus from (H) at $t = 1$, 5, and 12 s, with u^2 of a formaldehyde-fixed nucleolus in black. (**J**) Distribution of surface tension of nucleoli, γ. All distributions show static (blue), μg (green), and flow (red) conditions. *N* in the legend denotes number of nucleoli [(C) to (E), (G), and (J)] and nuclei (F) analyzed. Means ± SD and *P* values are shown in tables S11 and S12. Scale bar, 5 μm.

To simulate microgravity and controlled flows, we chose to use an RPM. While RPMs have been widely used for microgravity simulation over the past two decades, most studies used cell immunofluorescence imaging of fixed cells or biochemical essays after the microgravity exposure (*56*–*58*, *62*, *89*). Live-cell imaging and analysis of microgravity-exposed cells have been lagging behind (*53*–*55*), with no such studies focusing on the genome. Our goal was to fill this knowledge gap, although accomplishing it presents substantial challenges. Specifically, to perform a high-resolution confocal imaging, cells must be cultured on a glass-bottom dish, which has to be completely filled with media and sealed without air bubbles for an RPM experiment. Air bubbles add stress to cells, leading to apoptosis, but their removal from a solution is not trivial. We developed a sample preparation protocol (see Materials and Methods), where the sample filling and sealing occur while immersed in another fluid, leading to air bubble–free samples and thus removing their adverse effects.

Another major confounding variable in any simulated microgravity experiment is the effect of fluid flow due to rotation. In real microgravity, such flows are nonexistent. To decouple the effects of flows and microgravity in the RPM experiments, we carried out a systematic comparison of different RPM rotational paths [clinostat, 3D clinostat, horizontal fast flip, and linear sawtooth path (*40*, *51*, *61*)], while measuring acceleration and flows inside the sample. Informed by this survey, we designed an optimal μg path, which is a custom linear sawtooth path at 42° s^{-1}, where flows are minimized. The μg path achieves microgravity after ~3 hours. To study the effects of flows on cells, we use a fast-flipping horizontal flow at 1*g*, which we found to generate substantial flows inside the sample. We then compared our observations in μg and flow regimes against those for cells in a static sample dish under traditional cell culture conditions as well as cells in a static sample dish that was filled and sealed. We found the static and filled samples to be equivalent, with no observed differences, in contrast to the samples that experienced μg and flow regimes. Specifically, we find that cells in the flow condition are often isolated and elongated, with many cells detached, while cells in the μg condition are densely packed to each other. Neither of these regimes leads to formation of MCSs (Fig. 2), which were often

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observed in earlier RPM studies (*68*, *90*). Our results show that the combination of air bubble–free environment, slow rotational speed, and long cell adherence time before the experiment leads to a highly desired reduction in cell detachment and MCS formation.

We used our RPM to investigate the effects of microgravity and flows on the cell nucleus in live human cells. The nuclear area and volume increase after exposure to both stimulated microgravity and flows (Figs. 3 and 4). However, we find the nuclear height unchanged, indicating that nuclei are becoming larger rather than flattening (fig. S8). In addition, nuclear eccentricity increases in the flow condition, which is consistent with the cell's elongated shape that we observe. Despite the changes to nuclear shape and volume, we find the thickness and structure of the NE remaining unchanged (fig. S7). When we assess the NE dynamics by analyzing its shape fluctuations, we find an increased amplitude in the small wavelength fluctuations (large q) after μg exposure (Fig. 4H), which indicates a softening of the NE, as bending rigidity κ dominates the NE at high q, with $\left\langle u_q^2 \right\rangle \sim \kappa^{-1} q^{-4}$ (*69*, *91*). In contrast, large wavelength fluctuations (small q), which are dominated by the NE's lateral tension σ, with $\left\langle u_q^2 \right\rangle \sim \sigma^{-1} q^{-2}$, remain unchanged (*69*, *91*). Upon cells' formaldehyde fixation, low q fluctuations are not accessible due to the system's crosslinking, while high q fluctuations can still occur between crosslinks. The NE fluctuations in an unperturbed live cell have been shown to be driven by actin and microtubule activity at low q and thermally at high q (*69*). We find that, upon μg exposure, the actin filaments exhibit an aberrant organization with many adhesion foci, and the cytoplasm becomes smaller, which is consistent with microtubules becoming shorter and in line with previous studies (*71*–*73*). Despite these cytoskeletal changes, the NE fluctuations remain unaltered at low q in the μg regime. In contrast, cells exposed to flows are either isolated and elongated or residing in groups, with unchanged shape and cytoskeletal organization. This is consistent with an earlier report showing that cells within such groups are shielded from the shearing by the flows (*58*).

Next, we turn our attention to the genome. We find that the width of the chromatin density distribution slightly increases upon both μg and flow exposure, indicating an increase in chromatin heterogeneity (Fig. 5A). Using DCS, we perform a detailed analysis of chromatin dynamics across the entire nucleus and measure chromatin's local displacements, velocities, displacement correlations, as well as *MSND* and nucleus-wide spatial displacement autocorrelation function. We find that the chromatin dynamics remains largely unchanged in all aspects, including its micronscale coherence (*75*), after the cells experience μg or flows. Our data show that the genome, its organization and dynamics, is incredibly robust and seems unperturbed by the effect of flows or simulated microgravity. Gene expression was reported to change upon cell's exposure to simulated gravity (*43*–*46*, *92*). However, these changes include both up/down-regulation of many genes; thus, the overall transcriptional activity contributing to chromatin dynamics may remain, on average, unchanged (*8*). It is important to note that flows generated by RPM are felt by the entire cell on its surface, i.e., the plasma membrane. We do observe differences between cells in flows, where cells become elongated in the flow's direction, and in simulated microgravity, where cells are similar to the static case (Fig. 2). This is consistent with flows being negligible in simulated microgravity. In both the flow and simulated microgravity regimes, external flows are likely not directly felt by the genome, as they are screened by the plasma membrane, cytoplasm, and the NE. Shear stresses due to flows could be mechanotransduced through the cytoskeleton on the nucleus, which could lead to changes in gene expression of some genes (*93*). Such changes are likely small at the timescale of our experiments given the lack of changes in the genome's dynamics (*MSND*; Fig. 5H).

Mechanical stresses to the nucleus, such as cells moving through tight constrictions or micropipette aspiration, have been found to cause DNA damage in the genome (*15*, *94*–*96*). We find that while cells' exposure to flows leads to increased DNA damage, this is not the case for cells in simulated microgravity, where we minimized the flows. This agrees with earlier simulated μg studies on adherent cells (*97*), while nonadherent suspension cells recorded a DNA damage increase (*92*, *98*). We compared our results against samples, where we chemically induced DNA DSBs using NCS. The size of DSBs caused by μg and flows is larger than that of NCS-induced DSBs but comparable to the static control. In all studied conditions, chromatin compaction at DSBs is higher than that of random chromatin sites, agreeing with past findings of NCS-induced DSBs (*82*, *99*). The spatial distribution of DSBs across the nucleus in the static, μg, and flow regimes is comparable to that of random sites, while NCS-induced DSBs are preferentially at the nuclear periphery. This is consistent with NCS entering the nucleus through the NE (*82*). In contrast, μg and flow regimes affect the entire nucleus at once, leading to a random distribution of DSBs. DNA damage has been shown to eliminate chromatin's coherent motions (*75*), yet we still observe them upon flow-induced DNA damage. This suggests that such DNA damage may be low and nonperturbing to the nucleus-wide chromatin dynamics. Such damage may increase with a longer exposure to the flows.

Last, we examine the nucleoplasm in response to simulated μg and flows. This is particularly intriguing, as the nucleoplasmic organization is maintained through liquid-liquid phase separations (*29*, *85*). We focus on the nucleolus, the largest liquid condensate in the nucleus. We see the size of nucleoli increase after exposure to flows, but not to μg, while we found the nuclear volume and area to increase in both situations. Thus, we speculate that nucleolar size increases upon flow exposure due to fluid flows shearing the nucleus. We also find the nucleolar surface smoother and chromatin near it more heterogeneous in both μg and flow regimes, while the surface tension γ remains comparable to that of nucleoli under physiological conditions (*32*). This suggests that chromatin-nucleolus interface is affected by μg and flows, with small wavelength fluctuations being suppressed. These changes, however, are likely not due to changes in the transcriptional activity inside (RNA Pol I) or outside (RNA Pol II) the nucleolus, as the spatial distribution of both RNA Pol I and II remains unchanged upon μg and flow exposure. Again, while the overall RNA Pol I and II activity has not changed, changes to gene expression may have occurred (*20*, *43*).

In conclusion, our systematic comparison of mechanical perturbations in the form of simulated gravity and flows to live human cells shows that they both lead to significant changes in cell shape, cytoskeletal organization, as well as nuclear shape and size. Our study demonstrates the importance of minimizing flows, a confounding variable in the simulated microgravity experiments (*51*, *52*), by designing flow-free RPM rotational paths, such as our μg path. We find that the nucleus changes its shape and size in both conditions, but the genome, its organization and dynamics, is extremely robust against these mechanical perturbations. While the flows lead to DNA damage, simulated μg does not. The nucleolar surface becomes smoother,

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suppressing short wavelength fluctuations in both conditions. Overall, we find that 24-hour exposure of the nucleus to simulated microgravity leads to rather subtle changes, which however could amplify with time and then affect cell physiology. In summary, our study provides new biophysical insights into the nuclear and genome organization, highlighting its robustness to mechanical perturbations. Such knowledge offers a new perspective for the role of gravity in a cell.

MATERIALS AND METHODS

RPM design and construction

We designed an RPM inspired by existing blueprints (*100*), which we custom adjusted as follows: To accommodate cell culture experiments in the RPM, it was scaled to fit inside a cell incubator (HeraCell 170i, Thermo Fisher Scientific), with the width of 20″. The RPM consists of a base, inner and outer frames, and stepper motors (Nema 17, StepperOnline) that turn the frames (fig. S1). The motor for the inner frame is connected to a slip ring to allow the wiring to maintain connection during the rotation of the outer frame. The outer frame motor and slip ring are connected to the RPM through dedicated mounts (fig. S1). The frames and mounting pieces were designed in SolidWorks and 3D-printed with a Fortus 3D Printer (Stratasys) using standard ABS filament. For detailed construction guide and validation, see the Supplementary Materials and data S1 and S2.

Cell culture

HeLa (CCL2) cell lines stably expressing H2B-GFP, or H2B-GFP and 53BP1-mCherry, or H2B-GFP and NPM-mApple, were cultured according to American Type Culture Collection recommendations. Cells were cultured in a humidified, 5% CO_2 (vol/vol) atmosphere at 37°C in Gibco Dulbecco's modified Eagle's medium (DMEM) supplemented with 10% fetal bovine serum (vol/vol), penicillin (100 units/ml), streptomycin (100 μg/ml; Invitrogen), and Plasmocin Prophylactic (4.5 μg/ml; Invivogen). Cells were mycoplasma free, as determined by the Invivogen PlasmoTest (Invivogen). Forty-eight hours before imaging experiments, cells were plated on 35-mm MatTek dishes with glass bottom no. 1.5 (MatTek). Twenty-four hours before imaging, cells were loaded on the RPM as described below. Control cells remained in the incubator for 24 hours. Before imaging, cell medium was changed to the Gibco CO_2-independent medium supplemented with L-glutamine (Invitrogen). A MatTek dish containing cells was mounted on the microscope stage kept in a custom-built environmental chamber maintained at 37°C with 5% CO_2 (vol/vol) delivery throughout the experiment. For lamin visualization, the stable cell line HeLa H2B-GFP was transfected with LMNA-mCherry plasmid (Addgene #55068). Cells were plated on 35-mm MatTek dishes with glass bottom no. 1.5 (MatTek) 48 hours before imaging experiments and transiently transfected 30 hours before imaging using Lipofectamine 2000, following the manufacturer's protocol (Invitrogen). Cells were then loaded on the RPM 24 hours before imaging. For fixation experiments, cells were fixed on 35-mm MatTek dishes with glass bottom no. 1.5 (MatTek) using 3.7% formaldehyde in Gibco's phosphate-buffered saline (PBS) at room temperature for 40 min and then washed three times with PBS every 5 min. All experiments were performed at a minimum of three independent biological repeats.

RPM loading

To ensure that a cell culture dish is completely filled with media and contains no bubbles, a custom protocol was developed. Twenty-four hours after seeding cells on the MatTek dishes, five thin layers of parafilm were applied between the imaging dish and its cap. The imaging dish was then fully submerged into a large vessel filled with PBS at 37°C. While submersed, the cell media was replaced with the Gibco CO_2-independent medium supplemented with L-glutamine, followed by closing the cap on the dish tightly. Capping the dish while submersed is critical for elimination of air bubbles, as these will always be introduced by capping in open air. The dish was then removed from PBS, cleaned, sealed with another layer of parafilm on top of a thin coating of silicon grease (Haynes), and mounted on the RPM. This procedure can also be performed with standard T25 flasks (Falcon). pH and dissolved oxygen levels in the cell culture media inside the sealed dish were assessed after 24 hours (table S2) using a benchtop pH meter (Accumet AB150, Fisher) and a handheld O_2 sensor (RCYAGO), respectively.

Supplementary Materials

The PDF file includes:

Supplementary Methods
Supplementary Text
Figs. S1 to S15
Tables S1 to S12
Legends for data S1 and S2
References

Other Supplementary Material for this manuscript includes the following:

Data S1 and S2

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